Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. Civil, ECE – A & B
Course Code & Title: ORO551 Renewable Energy Sources
Name of the Faculty member (s): Mr.E. Thangam

Innovative Practice Description

- **Unit / Topic:** Unit II / Solar Thermal Collector, Evacuated Type Flat plate collectors
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO - 4
- **Activity Chosen:** Field Visit, Demonstration
- **Justification:**

The evacuated tube collector consists of a number of sealed glass tubes which have a thermally conductive copper rod or pipe inside allowing for much high thermal efficiency and working temperature compared to the flat plate solar collectors even during a freezing cold day. After teaching the constructional, I thought of conducting this as field visit for making the students to give overview of flat plate collector which enhance the learning level and as a teacher I can judge the understanding level of the students..

- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 35,

Photographer: one student – Mr. Kumaraguruparan (interested in photography)

Reporter: Myself

At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about convection mode of heat transfer in solar collector for 2 minutes.
- ✓ Then I told them to Pair with their neighbours and discuss about the operation of solar collector for another 1 minute.
- ✓ Finally, I randomly asked students about the operation of solar collector. (2 minutes)

- **CO – PO / PSO mapping:**

CO	PO 1	PO 7	PO 9	PO 10	PO12
CO 2	2	2	2	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 7**
	1
Justification for correlation	The basic applications of solar energy in an environmental context are taught to the students, hence it is moderately correlated (level 2).

- **Images / Screenshot of the practice:**

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***

  Students were satisfied with the field visit demonstration class on Solar thermal system, also they said, it increases our curiosity to explore more about the topic.

- ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society
2. Students understood the concept which was reflected from their answers for the questions I have asked during discussion session of field visit.
3. The success of the activity was evaluated by asking the same question during IAT-II – **Around 80% of students answered.**
4. Students can able to explain the concepts in examination without any confusion.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity, slow learners were not able to answer the question.

References:

✓ Rai G.D., “Non-Conventional Energy Sources”, Khanna Publishers, 2011

Signature of Faculty Member **HOD**

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Innovative Practice Description

- **Unit / Topic:** Unit III / Solar Photovoltaic Energy Conversion
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO - 9
- **Activity Chosen:** Field Visit, Demonstration
- **Justification:**

Solar Photovoltaic (PV) is a technology that converts sunlight (solar radiation) into direct current electricity by using semiconductors. When the sun hits the semiconductor within the PV cell, electrons are freed and form an electric current. Solar PV technology is generally employed on a panel (hence solar panels). After teaching the constructional, I thought of conducting this as field visit for making the students to give overview of PV panel which enhance the learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 35,

Photographer: one student – Mr. Kumaraguruparan (interested in photography)

Reporter: Myself

At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about solar PV system operation and its position for 2 minutes.
- ✓ Then I told them to Pair with their neighbours and discuss about the operation of solar photovoltaic energy conversion system for another 1 minute.
- ✓ Finally, I randomly asked students about the operation of solar photovoltaic energy conversion system. (2 minutes).

- **CO – PO / PSO mapping:**

CO	PO 1	PO 7	PO 9	PO 10	PO12
CO 3	2	2	2	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 7
	1
Justification for correlation	The basic applications of solar energy in an environmental context are taught to the students, hence it is moderately correlated (level 2).

- **Images / Screenshot of the practice:**

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***

  Students were satisfied with the field visit demonstration class on Solar PV system, also they said, it increases our curiosity to explore more about the topic.

- ***Benefit of the practice:***

1. Students can able to understand the impact of engineering solution on society
2. Students understood the concept which was reflected from their answers for the questions I have asked during discussion session of field visit.
3. The success of the activity was evaluated by asking the same question during IAT-II – **Around 80% of students answered.**
4. Students can able to explain the concepts in examination without any confusion.

- ***Challenges faced in implementation:***

1. Time utilization for conducting activity, slow learners were not able to answer the question.

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